

6.7 Membership and identity operators

in / not in / membership operators

The `in` and `not in` operators, known as membership operators, yield `True` if the left operand matches the value of an element in the right operand, which is a list or other iterable.

PARTICIPATION ACTIVITY

6.7.1: Membership operators: Checking for a value in a list.

Start☐ 2x speed

Python interpreter

```
>>> prices = ["$20", 15, 5]
>>> print (15 in prices)
True
>>> print (44 in prices)
False
```

Python (command line)

prices	"\$20"	15
	X	✓
	X	X

Captions ^

1. A user creates a new list.
2. Every element in the list is checked for the value of 15.
3. Every element in the list is checked for the value of 44, which is not found. So the statement is `False`.

Figure 6.7.1: Membership operators example: Checking for an item in a list.

```
# Use the "in" operator
barcelona_fc_roster = ["Alves", "Messi",
"Fabregas"]

name = input("Enter name to check: ")

if name in barcelona_fc_roster:
    print(f"Found {name} on the roster.")
else:
    print(f"Could not find {name} on the roster.")
```

```
Enter name to check: Messi
Found Messi on the roster.
...
Enter name to check: Rooney
Could not find Rooney on the roster.
```

```
# Use the "not in" operator
barcelona_fc_roster = ["Alves", "Messi",
"Fabregas"]

name = input("Enter name to check: ")

if name not in barcelona_fc_roster:
    print(f"Could not find {name} on the roster.")
else:
    print(f"Found {name} on the roster.")
```

```
Enter name to check: Messi
Found Messi on the roster.
...
Enter name to check: Rooney
Could not find Rooney on the roster.
```

substring

A substring, or matching subset of characters, of a larger string.

Figure 6.7.2: Checking for substrings.

```
request_str = "GET index.html HTTP/1.1"

if "/1.1" in request_str:
    print("HTTP protocol 1.1")

if "HTTPS" not in request_str:
    print("Unsecured connection")
```

```
HTTP protocol 1.1
Unsecured connection
```

Figure 6.7.3: Checking for membership in a dict.

```
my_dict = {"A": 1, "B": 2, "C": 3}

if "B" in my_dict:
    print("Found \"B\"")
else:
    print("\"B\" not found")

# Membership operator does not check values
if 3 in my_dict:
    print("Found 3")
else:
    print("3 not found")
```

Found
3 not

identity operator / is

The programmer can use the identity operator, `is`, to check whether a variable points to a single object.

is not

The inverse identity operator, `is not`, gives the negated value of `is`.

Figure 6.7.4: Identity operators.

```
w = 500
x = 500 + 500 # Create a new object with value 1000
y = w + w     # Create a second object with value 1000
z = x         # Bind z to the same object as x

if z is x:
    print("z and x are bound to the same object")
if z is not y:
    print("z and y are NOT bound to the same object")
```

z and x are bound
z and y are NOT

CHALLENGE ACTIVITY

6.7.1: Membership and Identity: Enter the output of the code.

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Start

Type the program's output

```
numbers = [1, 2, 3, 4]
x = int(input())

if x in numbers:
    print("in")
else:
    print("not in")
```

Input

5

Output

1

2

3

4

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ACTIVITY**

6.7.2: Boolean operators: Detect specific values.

Write an expression using membership operators that prints "Special number" if special_num is in the list special_list = [-99, 0, 44].

Sample output with input: 17

Not special number

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```
1 special_list = [-99, 0, 44]
2 special_num = int(input())
3
4 if """ Your solution goes here """ :
5     print("Special number")
6 else:
7     print("Not special number")
```

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